SS2P2: A Stable State-Space Point Process for Limit-Order-Book Simulation

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Abstract

Neural marked temporal point processes (MTPPs) predict the next limit-order-book (LOB) event well but *simulate* order flow poorly: closed-loop, their learned self-excitation drifts or explodes, and we show the standard remedy — rescaling the intensity to match the real event rate — often has *no solution* for unbounded models. We introduce SS2P2 (Stable State-Space Point Process), which keeps an expressive state-space backbone verbatim and replaces only the output heads: a *softmin-bounded rate head* whose intensity carries a hard closed-form ceiling (an exact dominating rate for thinning) with a floor of exactly zero, and a *rate-neutral* soft-max mark head that can be arbitrarily deep without perturbing the total rate. On a strict multi-seed benchmark over four crypto assets (Gemini ETH; Coinbase BTC/ETH/SOL, 6–13$\times$ faster) with lossless streaming evaluation and *verified* rate calibration for every model, SS2P2 is the best-calibrated simulator — best rate control and Fano fit on the home asset — and its prediction overtakes the strongest baseline (the neural Hawkes CT-LSTM) as streams grow faster, winning every prediction metric on SOL. Treating calibratability itself as an experimental outcome yields a dose–response: the unbounded parent head fails verified calibration on 8 of 12 checkpoints (all three on SOL), with a near-discontinuous rate response characteristic of criticality, while the bounded head fails once in twelve. The full recipe — bounded heads, stateful (TBPTT) training, $O(1)$ carried-state roll-out, certificate-preserving post-hoc calibration — simulates at verified real rates with close-to-empirical clustering. A negative result completes the picture: the textbook unbiased Monte-Carlo compensator collapses training under clipped adaptive optimization — estimator variance, not bias, governs whether likelihood-based rate calibration can be learned. All tables and figures are generated by released scripts.

Introduction

The limit order book (LOB) is the core mechanism of electronic markets: a continuous-time stream of order placements, cancellations, and trades whose fine structure drives price discovery and liquidity. Two tasks are asked of a generative model of this stream. The first is *prediction*: given the event history, score the time and type of the next event. The second is *simulation*: roll the model forward freely, feeding its own samples back as history, and reproduce the stylized facts of real order flow — calibrated event rates, burst–lull clustering, and heavy-tailed returns (Cont 2001). Simulation is the harder and, for downstream use (market-making world models, agent training, counterfactual analysis), the more valuable of the two (Li et al. 2025; Nagy et al. 2025).

Neural MTPPs (Du et al. 2016; Mei and Eisner 2017; Shchur et al. 2021) excel at prediction, and continuous-time state-space variants (Chang et al. 2025; Shi and Cartlidge 2022) push held-out likelihood to the state of the art. But that literature evaluates almost exclusively teacher-forced, and the very self-excitation that wins the one-step likelihood becomes a liability closed-loop: each sampled event raises the intensity, which produces more events, and with no mechanism bounding the rate the roll-out drifts hot or explodes. The standard patch — rescale the intensity until the simulated rate matches reality — turns out to be unavailable exactly where it is needed most: we find trained unbounded state-space checkpoints whose closed-loop rate is *near-discontinuous* in the rescaling constant (a 10^{-3} change flips the rate across the entire tolerance band) so that no constant rescaling exists, and the failure worsens systematically with the event rate of the asset. The field thus faces an *expressivity–stability frontier*: expressive intensities predict but cannot be trusted (or even calibrated) closed-loop, while trivially stable samplers simulate innocuously but predict poorly.

SS2P2 is a head transplant, not a new backbone. We keep the stacked latent-linear-Hawkes state-space backbone verbatim — its diagonal zero-order-hold dynamics, parallel-scan computability, and LayerNorm'd stack output $u(t)$ — and replace only what comes after $u(t)$:

- a **softmin-bounded rate head**: a gated bounded state $h = \sigma(W_o u) \odot \tanh(u)$ is read out through a smooth one-sided cap $z = c - \mathrm{softplus}(c - w^\top h - b)$, so $\lambda = s \cdot \mathrm{softplus}(z) \leq s \cdot \mathrm{softplus}(c)$ is a *hard closed-form ceiling* — an exact dominating rate for Ogata thinning — while $z \to z_{\text{raw}}$ as $z_{\text{raw}} \to -\infty$, so the floor is exactly zero and log-intensity keeps unit gradient in the quiet regime;
- a **rate-neutral mark head**: $p^*(k \mid t) = \mathrm{softmax}(\mathrm{MLP}(u))_k$ lives on the probability simplex, so $\sum_k \lambda_k = \lambda$ holds regardless of how nonlinear the mark network is — expressiveness goes where it cannot destabilize the rate.

Because the backbone is identical, differences against the unbounded parent (S2P2) are attributable to the heads. Our contributions:

1. **The SS2P2 head pair** (softmin-bounded rate $\times$ rate-neutral marks): a provably bounded, thinning-exact intensity on an unmodified state-space backbone, trained with stateful (TBPTT) likelihood and simulated with $O(1)$-per-event carried state.
2. **A strict two-axis benchmark** on four crypto assets (Gemini ETH; Coinbase BTC/ETH/SOL) $\times$ six models $\times$ three training seeds: prediction scored losslessly on *every* test event by a streaming evaluator with carried state, simulation scored on calibrated carried-state roll-outs against equal-duration real segments, checkpoint-level confidence intervals throughout. SS2P2 is the best-calibrated simulator on the home asset and *overtakes* the best predictor (NHP) as streams grow faster, winning every prediction metric on SOL. A per-gap decomposition explains the remaining prediction gap mechanistically: it is quiet-regime hazard, the deliberate price of the bound.
3. **Calibratability as a measurable model property.** Every model is rate-calibrated by a mark-preserving bisection with *full-scale verification*, and the outcome is reported as a result: the unbounded head fails verified calibration on 8/12 checkpoints with a critical-point signature, degrading with event rate (all three seeds on SOL); the bounded head fails 1/12. To our knowledge this is the first cross-asset quantification of closed-loop calibratability for neural TPPs.
4. **Exact post-hoc rate calibration** for factorized heads: the rate is linear in a learnable scale, the thinning ceiling scales with it, and the mark distribution is untouched — a certificate-preserving one-dimensional root-find that also *improves* every structure fact by repairing clock-rate distortion.
5. **A cautionary negative result on unbiased compensators.** The global Monte-Carlo compensator, though verifiably unbiased, collapses training under gradient clipping and adaptive optimization. Estimator variance, not bias alone, decides whether likelihood-level rate calibration succeeds.

All tables and figures in this paper are generated end-to-end by released collection and plotting scripts.

Background

Marked temporal point processes. An MTPP models a sequence $\mathcal{H}_T = \{(t_i, k_i)\}_{i=1}^{N_T}$ of events with times t_i and types $k_i \in \{1, \dots, K\}$ through per-type conditional intensities $\lambda_k^*(t)$. Writing $\lambda^*(t) = \sum_k \lambda_k^*(t)$ for the total (ground) intensity and $p^*(k \mid t) = \lambda_k^*(t)/\lambda^*(t)$ for the mark distribution, the log-likelihood of a sequence is

$$\log \mathcal{L} = \sum_i \Big[\log \lambda^*(t_i) + \log p^*(k_i \mid t_i)\Big] - \int_0^T \lambda^*(s)\, ds,$$

the integral being the *compensator* $\Lambda^*(t) = \int_0^t \lambda^*(s)\, ds$. By the random time change theorem, the rescaled masses $u_i = \Lambda^*(t_i)$, $\Delta u_i = u_i - u_{i-1}$ are i.i.d. $\mathrm{Exp}(1)$ iff the intensity is correctly specified — we report their mean (target 1) and KS distance as timing-calibration diagnostics. Sampling uses Ogata thinning, which requires a dominating rate $\bar{\lambda} \geq \lambda^*(t)$; a model whose intensity has no computable upper bound must resort to heuristic (and bias-prone) dominating-rate estimates. Classical self-exciting intensities are Hawkes processes (Hawkes 1971), widely used for LOB event flow (Jain et al. 2024a; Morariu-Patrichi and Pakkanen 2022); neural versions replace the parametric kernel with recurrent (Du et al. 2016; Mei and Eisner 2017) or attention-based (Zhang et al. 2020) history encoders.

The S2P2 backbone. S2P2 (Chang et al. 2025; Shi and Cartlidge 2022) is a state-space point process: L stacked latent-linear-Hawkes/SSM layers with diagonal state, ZOH-discretized between events and computable by parallel scan,

$$x_{t'}^{(\ell)} = \bar{A}^{(\ell)} x_t^{(\ell)} + (\bar{A}^{(\ell)} - I)\, \tilde{B}^{(\ell)} u^{(\ell-1)}\ \big[+ \tilde{E}^{(\ell)} \alpha \text{ at an event}\big],$$

with a LayerNorm'd residual readout $u^{(\ell)} = \mathrm{LayerNorm}\big(\mathrm{GELU}(y^{(\ell)}) + u^{(\ell-1)}\big)$ producing the stack output $u(t) := u^{(L)}(t) \in \mathbb{R}^H$. The published model reads the rate from $u(t)$ through a projection-and-softplus head, $\lambda = s \cdot \mathrm{softplus}(w^\top u + b)$, which is unbounded above; marks are coupled to the same rate field. SS2P2 keeps everything up to and including $u(t)$ and changes only these two heads.

Evaluation axes. We score *prediction* per genuine test event — the time/mark NLL split, type accuracy and perplexity, expected-time MAE, and the time-rescaling diagnostics above — and *simulation* by calibrated closed-loop roll-outs ($32 \times 600\,$s) bucketed into $1\,$s bins, compared to equal-duration bootstrap samples of real data on event rate, Fano factor across scales, volatility clustering ($|r|$-ACF), and return autocorrelation (Cont 2001; Vyetrenko et al. 2020).

Method: SS2P2

SS2P2 factors the per-type intensity into a bounded total rate and a rate-neutral mark distribution, both reading the *same* backbone embedding $u(t)$:

$$\lambda_k(t) \ = \ \lambda(t) \cdot p^*(k \mid t), \qquad \sum\nolimits_k \lambda_k(t) = \lambda(t),$$

the second identity holding because $\sum_k p^*(k \mid t) = 1$. The design principle is to put nonlinearity where it cannot perturb the rate.

Softmin-bounded rate head

From the LayerNorm'd (hence bounded-scale) embedding u, a gate produces a two-sided-bounded state, which is read out through a smooth one-sided cap:

$$\begin{aligned} o(u) &= \sigma(W_o u + b_o) \in (0,1)^H, \\ h(u) &= o(u) \odot \tanh(u) \in (-1,1)^H, \\ z_{\mathrm{raw}} &= w^\top h + b \ \text{ (unconstrained)}, \\ z &= c - \mathrm{softplus}\big(c - z_{\mathrm{raw}}\big), \\ \lambda(t) &= s \cdot \mathrm{softplus}(z), \end{aligned} \tag{1}$$

with a learnable positive scale s initialized so the baseline rate matches a target rate ($h \to 0 \Rightarrow \lambda \approx s \ln 2$).

Two properties follow immediately from Eq. (1):

- **Hard ceiling (simulation stability).** $z \leq c$ always, hence $\lambda(t) \leq s \cdot \mathrm{softplus}(c)$ in closed form. This is an *exact* dominating rate for Ogata thinning — the roll-out cannot run away, whatever state the closed loop visits.
- **Open floor (prediction honesty).** As $z_{\mathrm{raw}} \to -\infty$, $z \to z_{\mathrm{raw}}$, so $\lambda \sim s\, e^{z_{\mathrm{raw}}} \to 0$ and $\log \lambda$ keeps unit gradient in the quiet regime. The floor is exactly zero.

The asymmetry is the point. Our original G1 head bounded z on *both* sides (an ℓ_1-capped readout of the gated state), giving $\lambda \in (\ell_-, \ell_+)$ with $\ell_- \approx 0.36\,\mathrm{ev/s}$. The ceiling ℓ_+ buys stability, but the nonzero floor welds the quiet regime to the burst scale: in a quiet gap of length Δt the compensator bleeds $(\ell_- - \lambda^{\mathrm{true}})\Delta t$ nats that no parameter setting can recover. The requirement is asymmetric — ceiling for simulation, zero floor for prediction — and the softmin cap builds exactly that asymmetry into the nonlinearity (the ablation chain in the Experiments section quantifies the effect).

Rate-neutral mark head

Marks read the same u but never change the total rate:

$$p^*(k \mid t) = \mathrm{softmax}\big(\mathrm{MLP}(u)\big)_k, \qquad \lambda_k(t) = \lambda(t)\, p^*(k \mid t).$$

Because the soft-max lives on the simplex, the mark network can be arbitrarily deep — or later conditioned on exogenous state such as an agent's resting quotes — without touching λ. This decouples the two failure modes: rate stability is owned entirely by the bounded head, mark expressiveness entirely by the simplex.

Likelihood and sampling

Training maximizes the standard MTPP likelihood with the compensator evaluated on the training window; the mark term is a categorical cross-entropy on $p^*(k_i \mid t_i)$, decoupled from the rate term. Closed-loop sampling uses thinning with the exact dominating rate $s \cdot \mathrm{softplus}(c)$. An S4-style log-spaced initialization of the backbone decay spectrum (timescales 0.02–60 s) is kept as a training option and evaluated in the head-ablation chain.

Carried-state simulation

The standard roll-out protocol re-encodes a sliding window of the last W events from a zero state at every step. This is faithful to windowed training but has two costs: $O(W)$ work per simulated event, and a hard memory ceiling — nothing older than W events can influence the future, so long-memory stylized facts (the slow decay of $|r|$ autocorrelation) are *architecturally* unreachable regardless of the model. Because the backbone is a state-space recursion, both costs vanish: we carry the packed decoder state and advance it with only the new event, which is mathematically identical to encoding the entire history from the seed (10^{-5} agreement in float32) at $O(1)$ per event — 31× faster at W=1024 and memory-unbounded. Attention- and window-shaped baselines have no such Markov state and retain the sliding window.

Stateful (TBPTT) training

Windowed maximum likelihood encodes every training window from a cold initial state, so the model only ever trains in a "history just began" transient and MLE explains the observed rate with an inflated baseline — the free roll-out, whose state is warm, then double-counts. We train instead with truncated backpropagation through time: batch lanes walk contiguous stretches of the stream in order, and the decoder state at the end of window t is passed — *detached* — as the initial state of window $t{+}1$ (resets only at genuine stream boundaries, to the learned initial state). Gradients remain window-local; state values flow from the start of the stream. This makes the training regime consistent with carried-state simulation, at unchanged per-step cost; it requires non-overlapping windows (stride = window length), so our controls match that setting.

Post-hoc rate calibration

The factorization buys one more property: $\lambda = s \cdot \mathrm{softplus}(z)$ is *exactly linear* in the learnable scale s, the thinning ceiling $s \cdot \mathrm{softplus}(c)$ scales with it, and the mark head is untouched by construction. The free-running rate — which is monotone but nonlinear in s through the closed-loop feedback — can therefore be calibrated by a one-dimensional geometric bisection on a multiplier k: roll out briefly at $k\,s$, compare the realized rate to the target, and iterate ($\sim$6 probe roll-outs in practice). The result is a certificate-preserving, sampling-exact simulator at any prescribed mean rate. For *any* decoder, a constant k multiplying the sim-time intensity is still mark-preserving — type probabilities λ_k/λ are unchanged — so the same bisection applies to every baseline; what is lost off-family is only the exactness of the thinning bound. We make the procedure falsifiable: the target is the *validation*-split rate, the bisection must bracket and converge, and a full-scale validation roll-out must verify the target within tolerance (failures escalate to full-scale probes; unresolved failures exclude the checkpoint and are reported). Transfer to the test split is reported, never gated. We use calibration as a *post-hoc* correction for the residual rate inflation caused by the biased endpoint-rule compensator (the experiments quantify the alternative of fixing the estimator itself, and why its naive form fails) — and, in the experiments, as a measurement instrument in its own right.

Experiments

Protocol

Our primary benchmark is Gemini ETH-USD event-driven order flow (seven days, 62 event types, validation-split rate $\approx$ 4.2 events/s), with a cross-asset replication on Coinbase BTC/ETH/SOL (below). Six models — NHP (the Mei–Eisner CT-LSTM), an LSTM decoder and a SAHP-style causal-attention decoder (Zhang et al. 2020) (both *adapted generic backbones* over our shared input embedding, not paper-faithful re-implementations), PCT-LSTM (a per-type parallel CT-LSTM), S2P2, and SS2P2 — are trained under an identical configuration: sequence length 1024, non-overlapping windows, identical gradient-step counts, three training seeds each. The protocol is strict end-to-end:

- **Prediction** is scored on *every* genuine test event by a lossless streaming evaluator: raw test segments are traversed in stream order with recurrent state carried across chunks (one cold start per genuine segment boundary); the evaluator hard-fails if any event goes unscored.
- **Simulation** uses carried-state closed-loop roll-outs ($32 \times 600\,$s, three roll-out seeds per checkpoint) compared to equal-duration bootstrap samples of real segments.
- **Every model is rate-calibrated** before scoring: a one-dimensional bisection on a time-rescaling constant k (mark-preserving for every decoder; see Method) targets the *validation*-split rate, then a full-scale validation roll-out must verify the target within 15%; failures escalate to full-scale probes and, if still unmet, the checkpoint is excluded and the failure reported as a result. SAHP is the exception: its closed-loop rate is unstable across warm-starts and no constant k transfers, so it is reported uncalibrated (k=1, marked †).
- **Statistics**: roll-out seeds are averaged within each checkpoint; 95% CIs are t-based across checkpoints. Any batch failure or non-finite value aborts the run.

Prediction, and where its nats live

Table 1: NHP wins one-step prediction; SS2P2 is the strongest of the rest on marks (accuracy 0.332 vs. 0.346, perplexity 10.47 vs. 10.26) and concedes mainly timing likelihood. A per-gap decomposition of timeNLL $= -\log \lambda^*(t_i) + \Delta u_i$ (event term + compensator mass) locates the concession precisely. On matched test windows, event terms are nearly equal (-2.60 NHP vs. -2.53 SS2P2); the gap is compensator mass accrued *late in long gaps*: on gaps $>2\,$s, NHP integrates 4.4 units of hazard where SS2P2 integrates 7.6 and S2P2 10.4. Figure 1 shows why: NHP's per-dimension decay toward a *learned steady state* tracks the falling empirical hazard of real gaps from $\sim$0.5 s onward and floors at 0.16 ev/s (real: 0.07), while the state-space family — whose quiet-state behaviour is an emergent readout of decaying latents with no dedicated resting-rate parameter — stays 2–4× hot at every gap age. A second, smaller cost is the ceiling itself: SS2P2's intensity sits within 5% of its bound on 47% of events (NHP: 17% of its natural saturation), giving up $\sim$0.15 nats of burst sharpness. Both costs are the deliberate price of the certificate; what the certificate buys is the subject of the next two subsections.

Simulation

Table 2 scores calibrated closed-loop roll-outs. SS2P2 posts the best rate control (rel-err 0.163) and the best Fano fit (0.426 ± 0.183; PCT-LSTM comparable at 0.408 ± 0.021), and Figure 2 shows its Fano-vs-scale *curve* rising with aggregation scale like real data — the signature of burst–lull clustering that short-memory simulators flatten. Honest caveats are in the caption: one SS2P2 checkpoint over-clusters (its two siblings lead both long-memory columns),

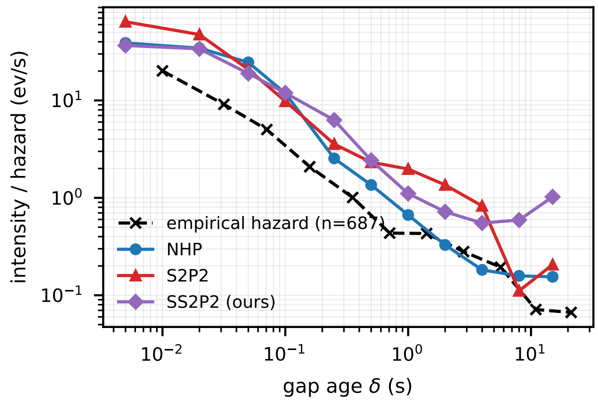

Figure 1: Mean model intensity at gap age δ, among test gaps that survive past δ, vs. the empirical hazard of the real gap distribution (log-log; survivor counts are small beyond 8 s). NHP tracks the falling hazard and floors near it; the state-space family stays hot through quiet gaps — the entire timeNLL gap of Table 1 accrues here.

and S2P2's surviving n=2 checkpoints carry a within-checkpoint roll-out variance an order of magnitude above every other calibrated model (MC sd 0.81 vs. $\leq$ 0.11 on simulated rate) — the same near-criticality that the next subsection makes a first-class result. NHP, the best predictor, is mid-pack in simulation: prediction and simulation quality remain distinct axes, and no unbounded model wins both.

Calibratability is a model property

Post-hoc rate calibration (Method) is mark-preserving for *every* decoder in the zoo — a constant k rescales sim-time intensity without touching type probabilities — so we calibrate every model, and treat the *outcome* of calibration as an experimental result in its own right. The protocol is deliberately falsifiable: probes must bracket and converge to the validation target, and a full-scale validation roll-out must verify it; every failure is reproducible (probe seeds are deterministic, and each retried failure reproduced its bisection bracket bit-identically).

Table 3 is, to our knowledge, the first cross-asset quantification of closed-loop calibratability for neural TPPs, and it shows a dose–response: at Gemini's $\sim$4 ev/s, one of three S2P2 checkpoints fails; at Coinbase rates (24–48 ev/s), seven of nine fail, leaving no calibrated S2P2 simulation arm at all on SOL. Figure 3 shows the failure signature: the closed-loop rate response of a failing S2P2 checkpoint is near-discontinuous in k — a 10^{-3} change in k flips the realized rate across the entire tolerance band (e.g. $36 \rightarrow 48.5\,$ev/s), and seed-to-seed variance at fixed k reaches $\sim$30% — the behaviour of a system parked near criticality, where no constant time-rescaling exists. The bounded SS2P2 response crosses the same target smoothly. This is the certificate made empirical: the same unbounded softplus head whose intensity-collapse states produce S2P2's pathological tMAE (Table 1) also parks checkpoints near criticality closed-loop,

model	overall↓	timeNLL↓	KS↓	mean u	tMAE(s)	ACC↑	PPL↓
NHP	**0.668±0.030**	**-1.660±0.010**	0.195±0.002	1.24±0.05	**0.29±0.02**	**0.346±0.009**	**10.26±0.28**
LSTM	1.814±0.136	-0.617±0.122	0.463±0.062	**1.16±0.53**	0.37±0.07	0.325±0.005	11.37±0.22
SAHP †	1.797±0.016	-0.663±0.024	0.511±0.006	0.75±0.06	0.45±0.05	0.309±0.003	11.70±0.10
PCT-LSTM	1.755±0.028	-0.905±0.029	0.380±0.030	1.26±0.14	0.49±0.10	0.231±0.005	14.30±0.11
S2P2	1.115±0.080	-1.279±0.078	**0.184±0.007**	1.84±0.02	9.11±17.42	0.321±0.000	10.96±0.02
SS2P2	1.229±0.341	-1.120±0.345	0.219±0.003	1.65±0.42	0.37±0.04	0.332±0.002	10.47±0.07

Table 1: Prediction on Gemini ETH per genuine test event (mean ± 95% CI over three training seeds; streaming lossless evaluator). NHP is the best predictor; SS2P2 is second on every mark metric and, unlike its unbounded parent S2P2, has no pathological expected-time tail (tMAE 9.1 ± 17.4 s for S2P2 — driven by intensity-collapse states — vs. 0.37 s).

model	rate_re↓	Fano_re↓	clus_re↓	rACF_re↓	cal. k
NHP	**0.126±0.098**	0.541±0.196	0.593±0.363	0.621±0.401	0.75±0.04
LSTM	0.268±0.141	0.895±0.102	0.942±0.280	0.827±0.253	0.40±0.22
SAHP †	0.588±0.352	0.731±0.322	0.829±0.229	0.661±0.172	1.00±0.00
PCT-LSTM	0.234±0.047	**0.408±0.021**	0.502±0.242	0.418±0.519	0.62±0.12
S2P2 (n=2)	0.258±1.835	1.377±8.547	**0.354±0.546**	**0.403±0.294**	0.60±0.37
SS2P2	0.163±0.176	0.426±0.183	1.998±6.900	2.959±11.602	0.47±0.17

Table 2: Closed-loop stylized facts on Gemini ETH (rel-err vs. real; roll-out seeds averaged per checkpoint, mean ± 95% CI over checkpoints). S2P2-s1 is excluded: it failed calibration verification (see *Calibratability*). † = SAHP uncalibrated by protocol. SS2P2 holds the best rate control and Fano fit; its clustering mean is inflated by one over-clustering checkpoint (s2: $|r|$-ACF 0.19–0.28 vs. real ∼0.02–0.06) — checkpoints s1/s3 lead the table outright (clus_re 0.46/0.33, rACF_re 0.34/0.19).

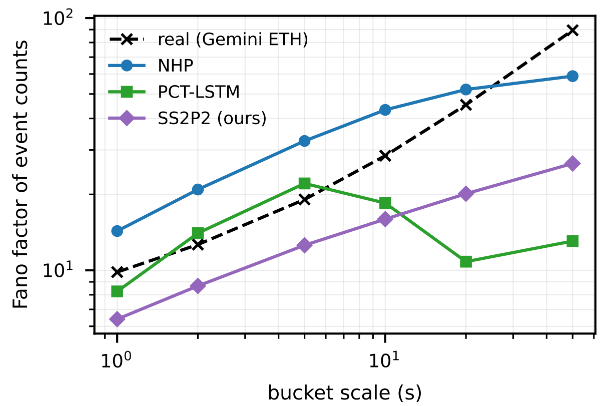

Figure 2: Fano factor of simulated event counts vs. bucket scale (Gemini ETH; mean over checkpoints and roll-out seeds, calibrated models). Real order flow is super-Poissonian with Fano *rising* in scale; SS2P2 reproduces the level and the rise most closely.

and both pathologies disappear under the softmin bound.

Cross-asset generalization

We replicate the full protocol on Coinbase BTC, ETH, and SOL (seven days; validation rates 38.3, 47.9, 24.0 ev/s — 6–13× Gemini, so the fixed 1024-event context spans only ∼20–40 s of market time). Table 4 reports prediction; Table 5 and Figure 4 simulation. Prediction *reverses* in SS2P2's favour as streams get faster and marks harder: on SOL it is best on every metric, including overall NLL (0.27 vs. NHP's 0.53), and on ETH it leads overall, KS, accuracy, and perplexity; on BTC NHP leads overall while SS2P2 ties it on marks. In simulation every calibrated model lands its rate (rel-err 0.06–0.30), Fano fits degrade for all models relative to Gemini (consistent with the shorter effective context), and SS2P2 and PCT-LSTM remain the strongest structure models. The stability findings replicate throughout (Table 3).

How the recipe was assembled: single-factor ablations

Three ablations on Gemini ETH assembled the SS2P2 configuration; each moves exactly one factor.

Carried-state roll-out unlocks long memory. Sliding-window simulation caps memory at the window span (∼27 s of market time at 64 events), making the slow decay of $|r|$-autocorrelation *architecturally* unreachable; with $O(1)$-per-event carried state (exactly equivalent to encoding the full history for state-space recursions), simulated $|r|$-ACF reaches real levels for the first time in this study. Memory truncation, not model class, was the binding constraint.

TBPTT owns structure and prediction — not the rate. Pairing cold-start against stateful training, all else fixed: prediction improves on every metric (S2P2's expected-time MAE collapses $15.6 \rightarrow 2.7$ s) and simulated clustering reaches near-real levels, while the free-running rate and per-gap compensator mass barely move (mean u 1.9–2.2 throughout). The residual rate inflation is a property of the *loss*, not of the state regime.

Calibration closes the rate and repairs structure. Applying the k-calibration to trained SS2P2 checkpoints cuts the closed-loop rate from ∼7× real to within tolerance and *improves every structure fact* (clustering rel-err $8.2 \rightarrow 0.45$ on the TBPTT arm): facts computed on an over-hot stream have mis-scaled buckets, so much of the apparent structure deficit of uncalibrated simulators is a clock-rate artifact. Calibration and stateful training compose: the TBPTT advantage persists after calibration.

model	Gemini ETH ∼4 ev/s	CB BTC 38 ev/s	CB ETH 48 ev/s	CB SOL 24 ev/s
NHP	3/3	3/3	3/3	3/3
LSTM	3/3	3/3	3/3	3/3
SAHP †	*model-level divergence: reported uncalibrated*			
PCT-LSTM	3/3	3/3	3/3	3/3
S2P2	**2/3**	**1/3**	**1/3**	**0/3**
SS2P2	3/3	**2/3**	3/3	3/3

Table 3: Checkpoints passing calibration + full-scale verification, out of three training seeds per dataset (bold marks failures). The unbounded S2P2 head degrades with event rate — 8 of 12 checkpoints uncalibratable overall, including *all three* on SOL — while the bounded SS2P2 head fails once in twelve. SAHP diverges at the model level on every asset.

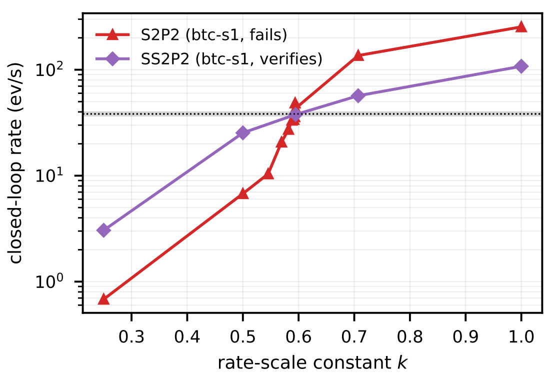

Figure 3: Calibration probe ladders on Coinbase BTC (target band shaded): the failing S2P2 checkpoint's closed-loop rate is near-discontinuous in the rescaling constant k; the verified SS2P2 checkpoint crosses the target smoothly.

A negative result: the unbiased compensator fails dynamically. Replacing the endpoint-rule compensator with the global Monte-Carlo estimator (verifiably unbiased; its expectation matches the model's true intensity mass to $6.4\times$ the endpoint value on a trained control) *collapses* training: free-running rates fall to 0.05–0.08 ev/s and teacher-forced NLL degrades by >5 nats while the training loss decreases smoothly. The estimator's heavy-tailed penalty is systematically trimmed by gradient clipping under Adam, and maximum likelihood learns to exploit the sparse audit — spiky intensity at events, crushed baseline elsewhere. Unbiased in expectation is not unbiased under clipped adaptive optimization; a stratified per-gap estimator is the indicated repair and left to future work.

Related Work

Neural TPPs for event streams. Neural MTPPs learn history-dependent intensities with recurrent (Du et al. 2016; Mei and Eisner 2017), attention-based (Zhang et al. 2020; Zuo et al. 2020), jump-diffusion (Zhang et al. 2024), and continuous-time state-space (Chang et al. 2025) encoders; see Shchur et al. (2021) for a survey. Shi and Cartlidge (2022) specialize neural Hawkes models to LOB event streams and evaluate both prediction and simulation. Notably, the state-space line is evaluated almost exclusively teacher-forced: Chang et al. (2025) report state-of-the-art held-out likelihood but never roll the model out (they avoid thinning altogether). Our contribution is orthogonal to the encoder line — we keep the state-space backbone verbatim and change what is read *out* of it, showing that the heads, not the backbone, own closed-loop stability and calibratability.

Hawkes modeling of order flow. Parametric Hawkes processes are the classical LOB event model (Hawkes 1971, 2018; Jain et al. 2024a,b), with state-dependent extensions (Morariu-Patrichi and Pakkanen 2022). Their linear structure gives exact stationarity and branching-ratio certificates but limited expressiveness. SS2P2 takes the complementary route: full neural expressiveness with stability restored by a closed-form intensity bound rather than by linearity.

Generative LOB simulators. Learned order-flow generators span GANs (Li et al. 2020; Coletta et al. 2021, 2022), token-level autoregressive state-space models (Nagy et al. 2023), diffusion-guided agents (Huang et al. 2026), and foundation-model-scale transformers (Li et al. 2025; Sood et al. 2026); agent-based simulators (Byrd, Hybinette, and Balch 2020; Frey et al. 2023) remain the RL workhorse. These systems are scored on distributional realism — stylized facts (Cont 2001; Vyetrenko et al. 2020) and, increasingly, the KS/Wasserstein suites of LOB-Bench (Nagy et al. 2025). Two properties separate SS2P2 from this line: a single likelihood-based model is held to *both* axes (per-event predictive NLL and closed-loop realism), and the simulated rate is not merely measured but *calibrated and verified*, with failures reported as first-class results. None of the generative simulators above carries a closed-loop stability certificate; TradeFM's inter-arrival marginal is its weakest statistic (Sood et al. 2026), and LOB-Bench's authors observe error accumulation over long autoregressive roll-outs (Nagy et al. 2025) — precisely the regime our bounded head and rate verification target.

Bounded intensities. Bounding or normalizing intensities appears in the TPP literature mainly as a computational device (dominating rates for thinning). Here the bound is a modeling decision with an asymmetry requirement — a hard ceiling for closed-loop stability, an exactly-zero floor for quiet-regime likelihood — and the softmin cap realizes precisely this pair. Our cross-asset calibration study makes the case empirical: unbounded softplus heads become progressively uncalibratable as the event rate grows, while the bounded head admits a verified constant-rate rescaling almost everywhere.

Conclusion

We held six neural MTPPs to both sides of the prediction–simulation tension on four crypto assets, under a protocol strict enough to falsify its own claims: lossless streaming prediction, multi-seed checkpoint-level confidence intervals, and verified rate calibration whose failures are reported as

model	BTC		ETH		SOL	
	overall↓	ACC↑	overall↓	ACC↑	overall↓	ACC↑
NHP	**-1.07±0.01**	**0.449±0.003**	-0.81±0.03	0.341±0.006	0.53±0.01	0.184±0.004
LSTM	-0.44±0.40	0.411±0.015	0.15±0.04	0.288±0.003	1.34±0.25	0.141±0.005
SAHP †	-0.93±0.01	0.412±0.001	-0.60±0.01	0.297±0.004	0.67±0.02	0.155±0.003
PCT-LSTM	-0.60±0.08	0.313±0.010	-0.34±0.10	0.248±0.007	0.89±0.03	0.105±0.004
S2P2	-0.40±0.23	0.439±0.003	-0.38±0.22	0.331±0.008	0.67±0.49	0.182±0.002
SS2P2	-0.87±0.11	0.447±0.003	**-0.87±0.08**	**0.345±0.003**	**0.27±0.05**	**0.192±0.001**

Table 4: Cross-asset prediction (mean ± 95% CI over training seeds). SS2P2 overtakes NHP as the event rate and mark entropy grow.

model	BTC		ETH		SOL	
	rate_re	Fano_re	rate_re	Fano_re	rate_re	Fano_re
NHP	0.09±0.08	0.86±0.07	0.37±0.20	0.87±0.11	**0.19±0.03**	0.75±0.17
LSTM	0.10±0.09	0.99±0.00	0.40±0.03	0.99±0.00	0.23±0.04	0.99±0.00
SAHP †	0.71±0.38	**0.76±0.20**	0.79±0.11	0.76±0.46	0.68±0.15	0.92±0.08
PCT-LSTM	**0.06±0.05**	0.79±0.12	0.42±0.26	**0.76±0.17**	0.22±0.24	**0.46±0.33**
S2P2	0.07	1.11	**0.27**	6.04	–	–
SS2P2	0.24±1.83	0.79±0.99	0.42±0.17	0.79±0.05	0.30±0.17	0.60±0.60

Table 5: Cross-asset simulation (rel-err vs. real). "–" = no calibrated checkpoints (all three S2P2 seeds failed verification on SOL).

results. The SS2P2 heads — a softmin-bounded rate with a closed-form thinning ceiling and an exactly-zero floor, times a rate-neutral simplex mark head on an unchanged state-space backbone — deliver the best-calibrated simulator on the home asset and overtake the strongest predictor as streams grow faster, while a per-gap hazard decomposition prices the residual prediction gap precisely where the design puts it: quiet-regime hazard, the cost of the bound. The sharpest finding is that *calibratability is a model property*: the unbounded parent head fails verified rate calibration on 8 of 12 checkpoints with a critical-point signature that worsens with event rate, while the bounded head fails once — the stability certificate made empirical. Stateful training and carried-state roll-out contribute structure that calibration alone cannot, and the unbiased Monte-Carlo compensator's collapse under clipped adaptive optimization warns that estimator variance, not bias, governs whether rate calibration can be *learned* rather than retrofitted. Because the mark head is rate-neutral, conditioning it on exogenous state — an agent's resting quotes — cannot destabilize the rate, making SS2P2 a suitable environment model for market-making world models; that action-conditioned extension, and a low-variance stratified compensator, are future work. All tables and figures are regenerated end-to-end by the released collection and display scripts.

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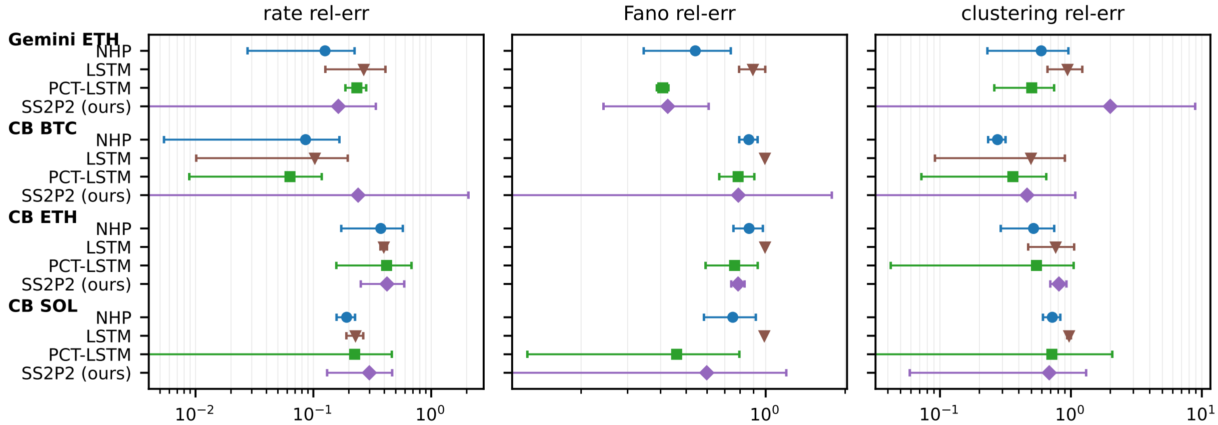

Figure 4: Cross-asset simulation quality (rel-err vs. real, log scale; whiskers = 95% CI over checkpoints). S2P2 and SAHP omitted where no calibrated checkpoints exist.

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